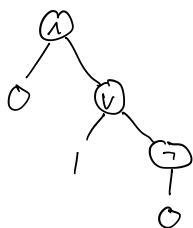
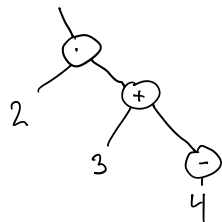


feks $2 \cdot (3 + (-4))$



funktioner

$$y = f(x_1, x_2) = 3x_1 + 5x_2 + 7$$

x_2	x_1	y
0	1	12
1	1	15
...	...	...

Boolske funktioner
med én input som: 1

x	$\text{not}(x)$
0	1
1	0

to input som: 11
10
01
00

Exercise I-1: Check Satisfiability

Which of the following formulas are satisfiable (give a satisfying assignment)? Which are not (give reasons)?

Note: in all exercises, logical negation (NOT) is denoted by "-".

- a) $A \wedge B$ ✓
- b) $A \vee B$ ✓
- c) $A \rightarrow B$ ✓
- d) $A \wedge \neg A$ ✗
- e) $A \vee \neg A$ ✓

Exercise I-2: Equivalent Formulas

Two formulas are equivalent, if the same assignments satisfy both of them.

Which of the following formulas are equivalent?

- a) $\neg A \wedge B$
- b) $\neg A \vee B$
- c) $A \rightarrow B$
- d) $(A \rightarrow B) \wedge (\neg B \rightarrow A)$
- e) $(\neg A \rightarrow B) \wedge (\neg B \rightarrow \neg A)$

A	B	$\neg A \wedge B$	$\neg A \vee B$	$A \rightarrow B$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	F	T	T

$(\neg A \vee B) \wedge (B \vee A)$	$(A \vee B) \wedge (B \vee \neg A)$
T	T
T	F
T	T
F	F

Exercise I-3: Convert to CNF

Convert the following formulas into CNF:

- a) $\neg A \wedge B$ $\neg A \wedge B$
- b) $\neg A \vee B$ $\neg A \vee B$

c) $A \rightarrow B \quad \neg A \vee B$

d) $(A \rightarrow B) \wedge (\neg B \rightarrow A) \quad (\neg A \wedge B) \wedge (B \vee A)$

Exercise II-1: Check Satisfiability

Which of the following formulas are satisfiable (give a satisfying assignment)? Which are not (give reasons)?

a) $(A \rightarrow B) \wedge (B \rightarrow A)$ ✓

b) $(A \rightarrow B) \wedge (B \rightarrow A) \wedge A$ ✓

c) $(A \rightarrow B) \wedge (B \rightarrow A) \wedge \neg A$ ✓

d) $(A \rightarrow B) \wedge (B \rightarrow \neg A) \wedge (\neg A \rightarrow \neg B) \wedge (\neg B \rightarrow A)$ ✗

Exercise II-2: Equivalent Formulas

Two formulas are equivalent, if the same assignments satisfy both of them.

Which of the following formulas are equivalent?

a) $(A \rightarrow B) \wedge (\neg B \rightarrow A)$

b) $(A \rightarrow \neg B) \wedge (B \rightarrow A)$

c) $(\neg A \vee \neg B) \wedge (A \vee \neg B)$

d) $(B \vee A) \wedge (\neg A \vee B)$

Exercise II-3: Convert to CNF

Convert the following formulas into CNF:

a) $(\neg A \rightarrow B) \wedge (\neg B \rightarrow \neg A) \quad (A \vee B) \wedge (B \vee \neg A)$

b) $A \rightarrow (\neg (B \wedge D)) \quad (\neg A \vee \neg B \vee \neg D)$

c) $A \rightarrow (\neg (B \vee D)) \quad (\neg A \vee \neg B) \wedge (\neg A \vee \neg D)$

d) $A \rightarrow (\neg (B \rightarrow (C \wedge D)))$
 $= A \rightarrow (\neg (B \rightarrow (C \wedge D)))$
 $= A \rightarrow (\neg (\neg B \vee (C \wedge D)))$
 $= A \rightarrow (B \wedge \neg (C \wedge D))$
 $= A \rightarrow (B \wedge (\neg C \vee \neg D))$
 $= \neg A \vee (B \wedge (\neg C \vee \neg D))$
 $= (\neg A \vee B) \wedge (A \vee (\neg C \vee \neg D))$